

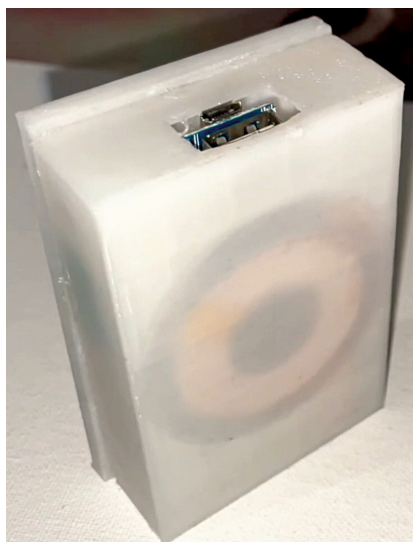


Fig. 15: The author's prototype



Fig. 16: Testing power bank and charging AirPods



Fig. 17: Testing power bank's wireless charging system by powering LED wirelessly

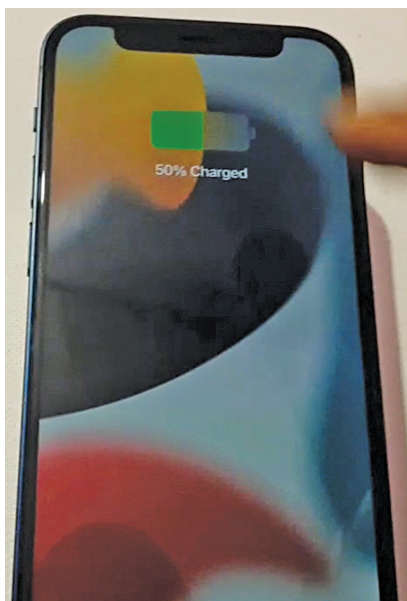


Fig. 18: A phone being charged

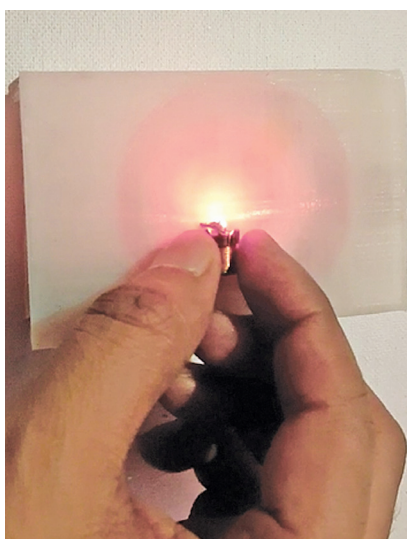


Fig. 19: LED glowing to indicate power transfer

either with the wires directly or a battery pack connector strip. In case of multiple batteries being used, make sure to connect the batteries in parallel (see Fig. 6). Next, connect the battery step-up charging module as shown in the circuit diagram. The wireless charging coil should now be soldered with the input of the wireless power transmitter module and its output with the charging module output pins (see Fig. 7 through Fig. 10).

The coil may now be fixed in the

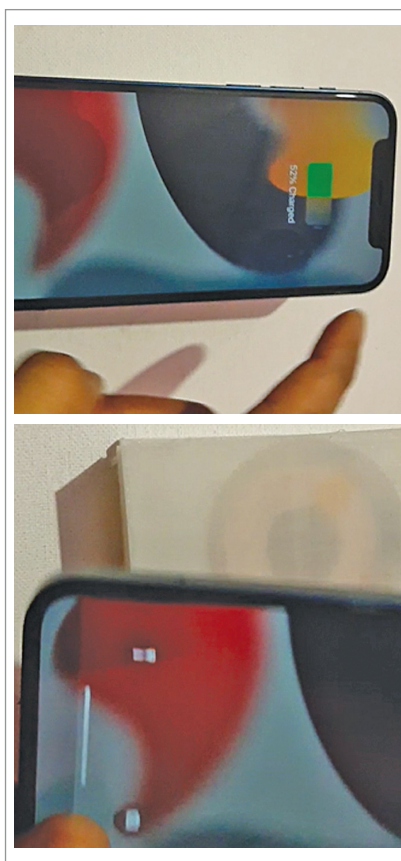


Fig. 20: Phone charging

cavity with a magnet for MagSafe. Above this fix the battery along with the BMS module in USB cavity of the 3D-printed case. Finally, fix the top cover (see Fig. 11 through Fig. 14). For MagSafe, fix the circular magnet in cavity with glue.

Testing

To test the circuit, plug in any device to the USB and notice it charging. (In the pictures an Airpod case is shown plugged in.) Next, bring any wireless device (a wireless LED is shown in the pictures) and see the LED glowing, as shown in Fig. 19, as the power bank emits power in the form of electromagnetic waves. A phone can be charged by placing it on the device, as shown in Fig. 20. **EFY**

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